

Standard Form & Indices

Answer sheet · 26 marks total

Section A — Laws of indices

A1. 4^8 (1)

A2. 7^7 (1)

A3. 6^6 (1)

A4. $2^3 \times 2^2 = 2^5$ (1) = **32** (1)

Section B — Zero and negative indices

B1. **1** (1)

B2. $1 \div 5^2 = 1 \div 25$ (1) = **1/25** (1)

B3. $1 \div 10^3 = 1 \div 1000$ (1) = **0.001** (1)

Section C — Writing standard form

C1. 7.2 (1) = **7.2×10^4** (1)

C2. 8.3 (1) = **8.3×10^6** (1)

C3. 4.8 (1) = **4.8×10^{-3}** (1)

C4. 3 (1) = **3×10^{-5}** (1)

Section D — Converting back, and calculating

D1. **65,000** (1)

D2. **0.00003** (1)

D3. $4 \times 2 = 8$, $10^3 \times 10^5 = 10^8$ (1) = 8×10^8 (1) already in standard form: **8×10^8** (1)

D4. $9 \div 3 = 3$, $10^8 \div 10^3 = 10^5$ (1) = 3×10^5 (1) already in standard form: **3×10^5** (1)

Marking note: award method marks for a correctly identified front number/power, or correctly applied law of indices, even if the final answer contains an arithmetic error. Accept equivalent fractions/decimals.